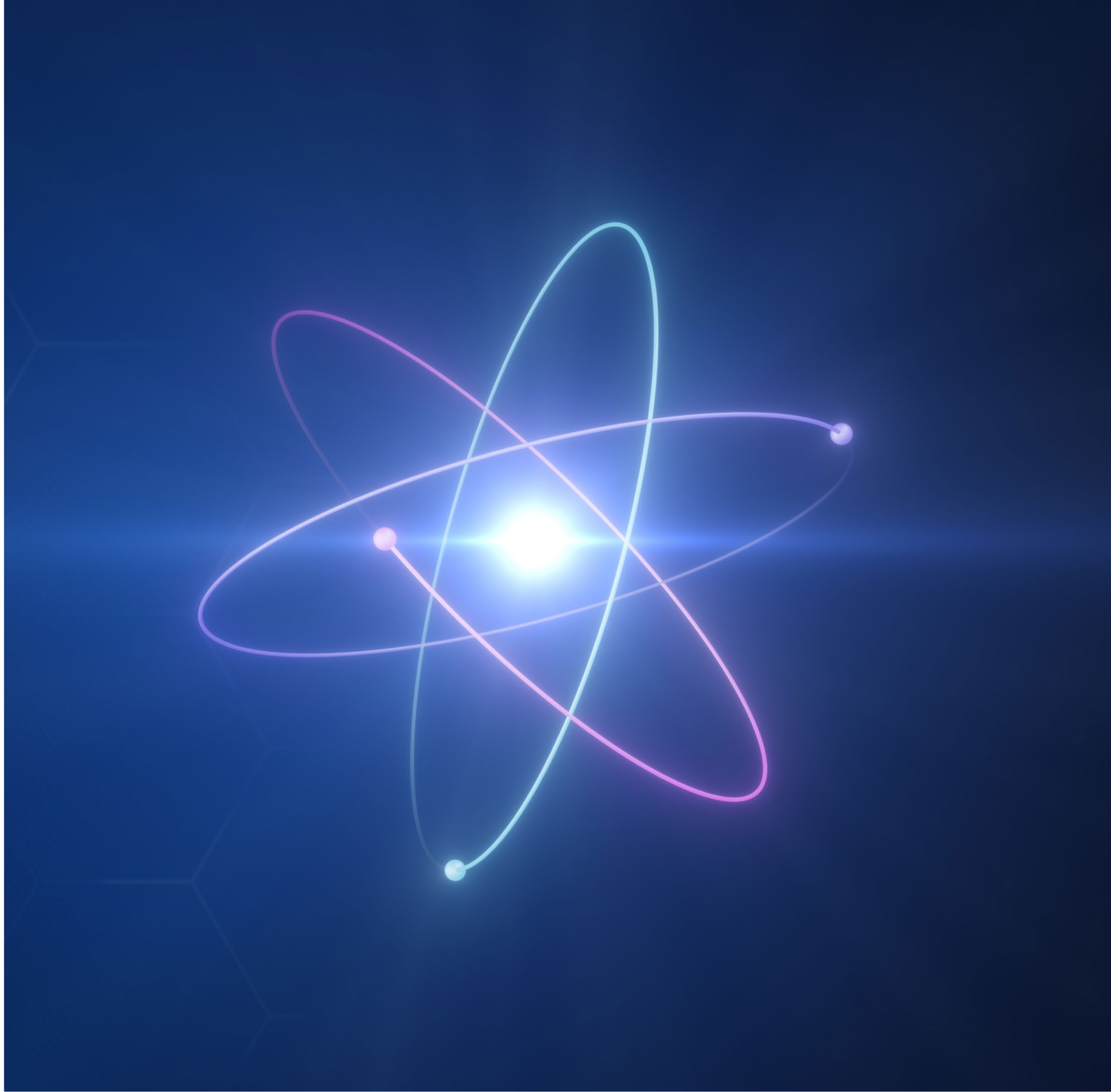
HSLU Lucerne University of Applied Sciences and Arts

Chemistry

SW02: Structure of atoms



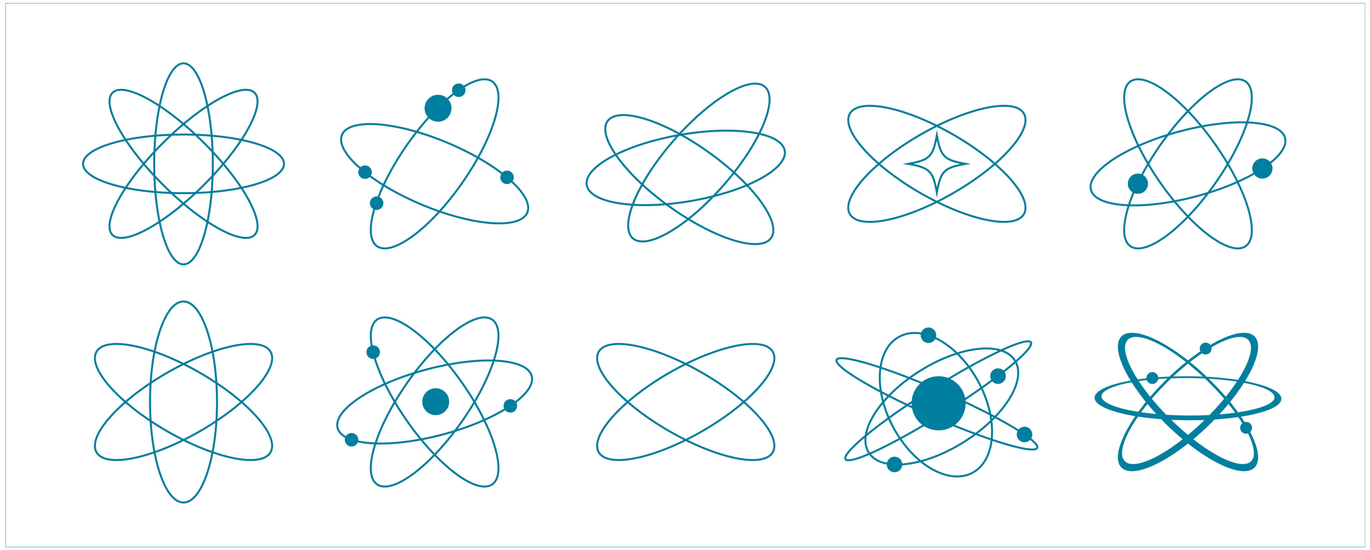
Dr. Macarena San Martin Ruiz

HS26

Engineering and Architecture

FH Zentralschweiz

Content



Structure of Atoms:

- Atomic nucleus, electron shell
- Chemical elements
- Isotopes
- Atomic mass (revision)
- Radioactivity



Learning objectives

Be able to read the following characteristics of a chemical element from the periodic table: **atomic number**, **mass number**, **number of protons**, **number of electrons**, **number of neutrons**, **relative atomic mass**

Be able to explain the terms **"nuclide"**, **"isotope"**, and **"isobars"**

Be able to describe **alpha**, **beta**, and **gamma decay** (what are the **products** and **reactants**, which **particles** are released?)

Recommended Chapters book Raymond Chang 14th edition:

Chapter 2:

More in detail 2.2, 2.3

Chapter 3:

More in detail 3.1

Chapter 19:

More in detail 19.1, 19.2, 19.3, 19.5, 19.6,

From Philosophy to Natural Science

Philosophers in the 6th century B.C. assumed that the material world was composed of the four elements: **earth, water, air, and fire**.

500 B.C.: *Atom* (= "indivisible"), **Leucippus and Democritus**

1661: Robert Boyle: Matter consists of small *"corpuscles"* (particles)

1805: John Dalton: First atomic model – each element consists of atoms that are characteristic of that element (in terms of mass, volume), **law of multiple proportions**





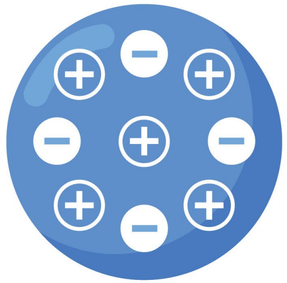
The Atomic Models

DALTON
1808



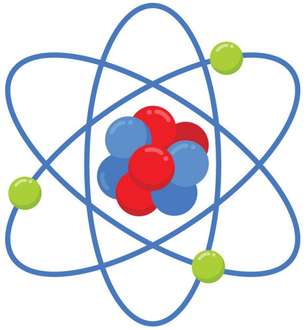
All matter is made of atoms

THOMSON
1897



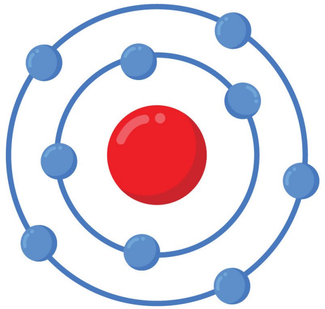
Discovered the inner structure of the atom

RUTHERFORD
1911



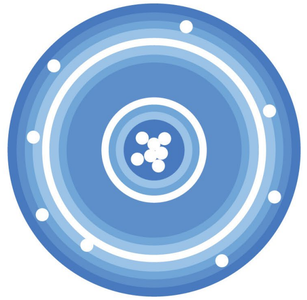
Discovered electrons surround the nucleus of an atom

BOHR
1913



Electron move is fixed orbitals (shells)

SCHRODINGER
1926



Wave behavior of the electron

Structure of Atoms

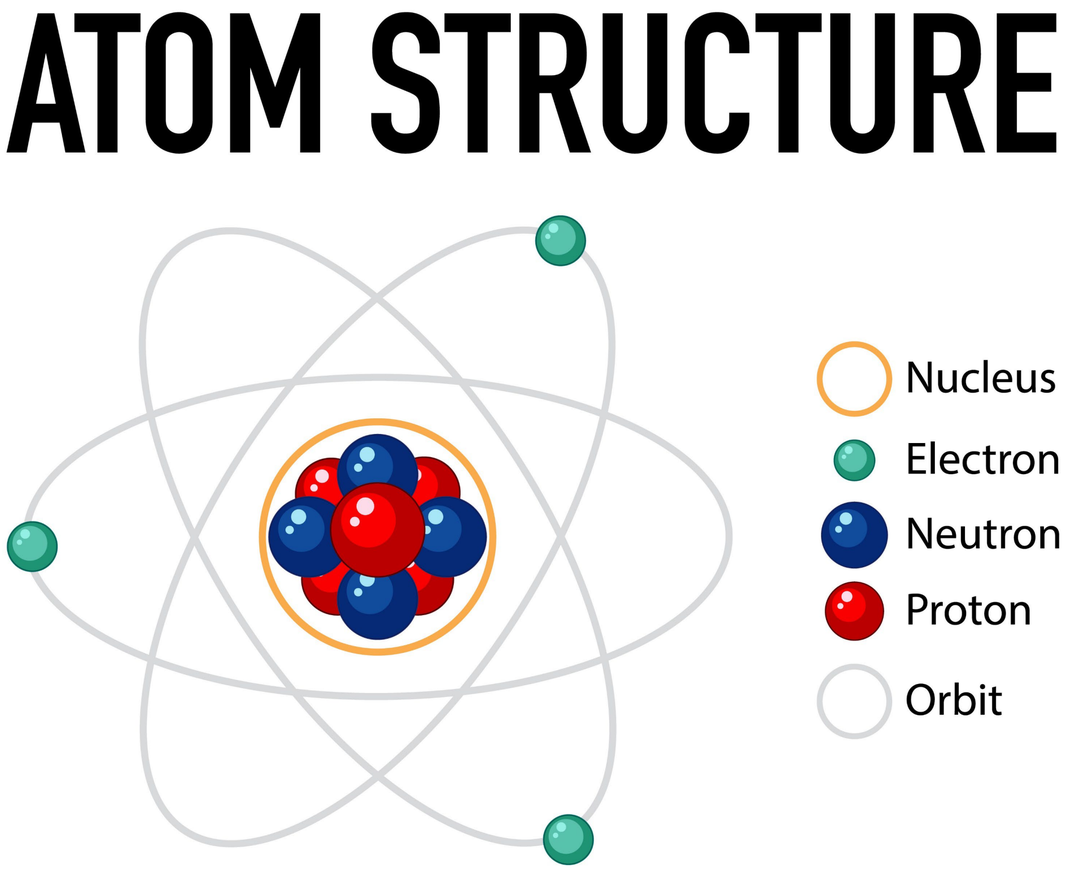
Atoms are the basic building blocks of matter
(*Greek: atomos = indivisible*)

Atoms are electrically neutral,

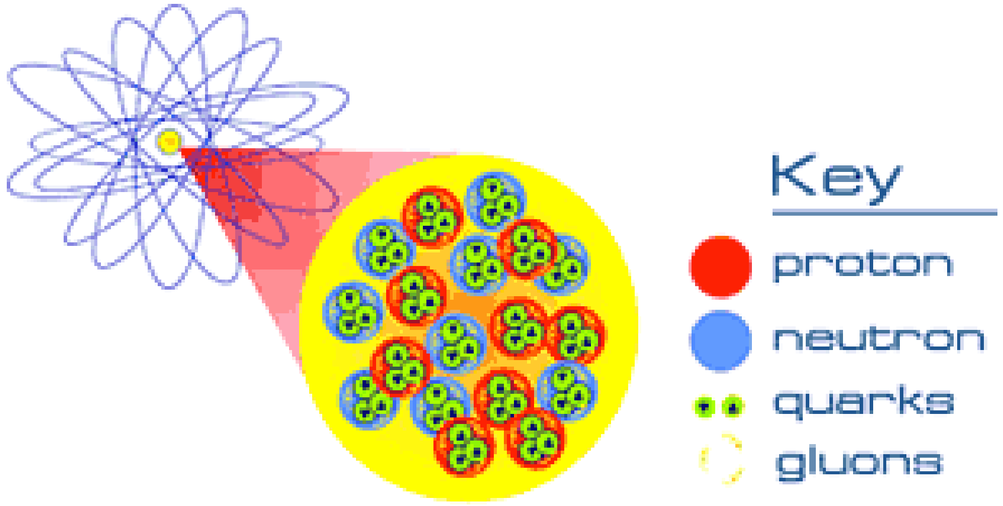
therefore:
Number of protons = number of electrons

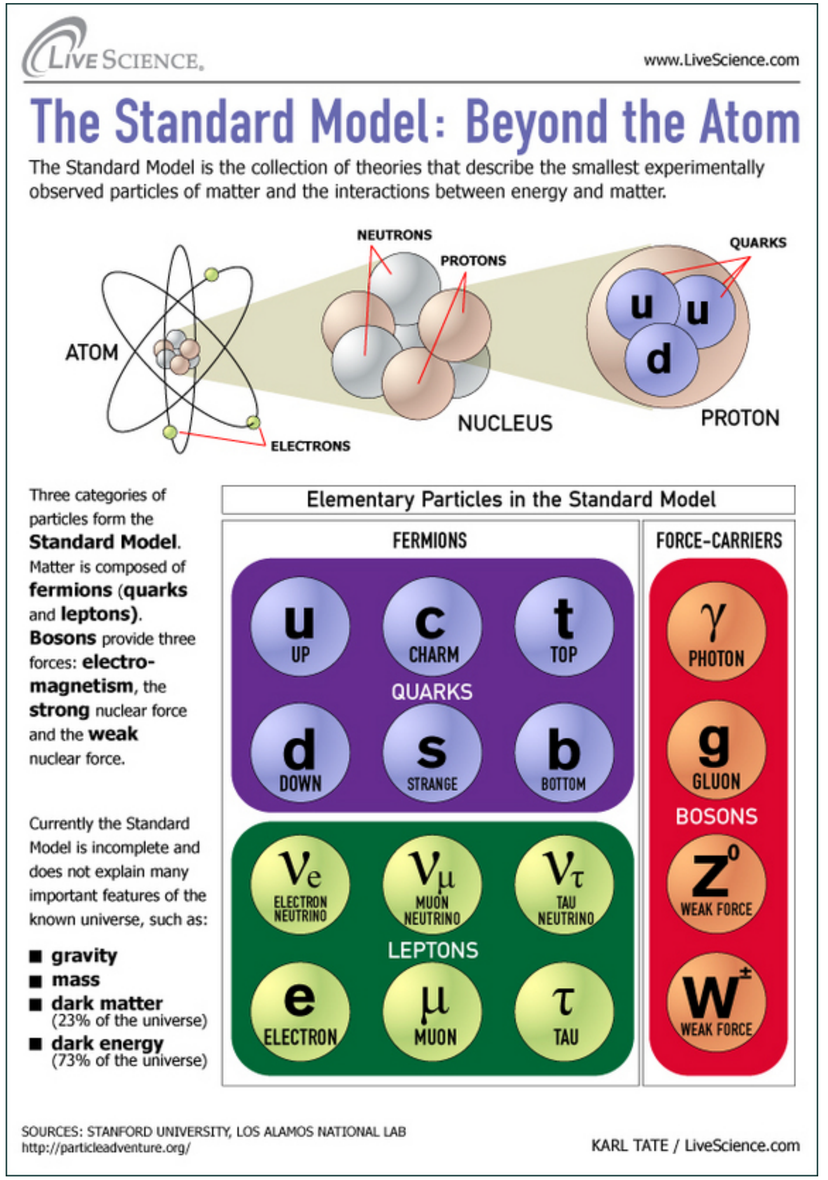
Subatomic particles: bosons, fermions, etc.

→ field of **particle physics**



Brief Excursus: Elementary Particles (not part of the curriculum)





In fact, the neutron and proton are themselves collections of smaller fundamental quarks.

In addition, there is a collection of bosons whose exchange mediates the four fundamental forces. γ, W+- , Z0, 8 gluons, graviton.

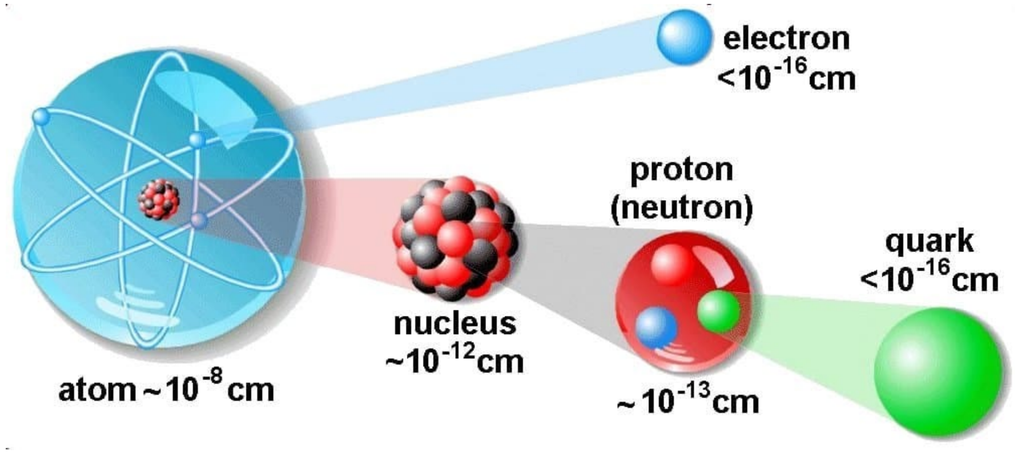
Only quarks and gluons experience the color force and quarks are never found in isolation.

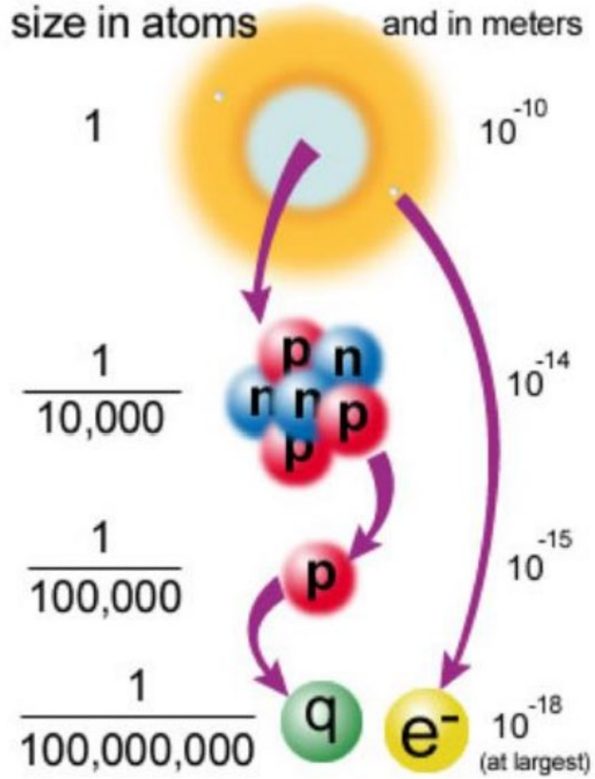
Size and Mass Ratios

$d_{atomic\ core} \approx 1/100{,}000\ d_{entire\ atom}$

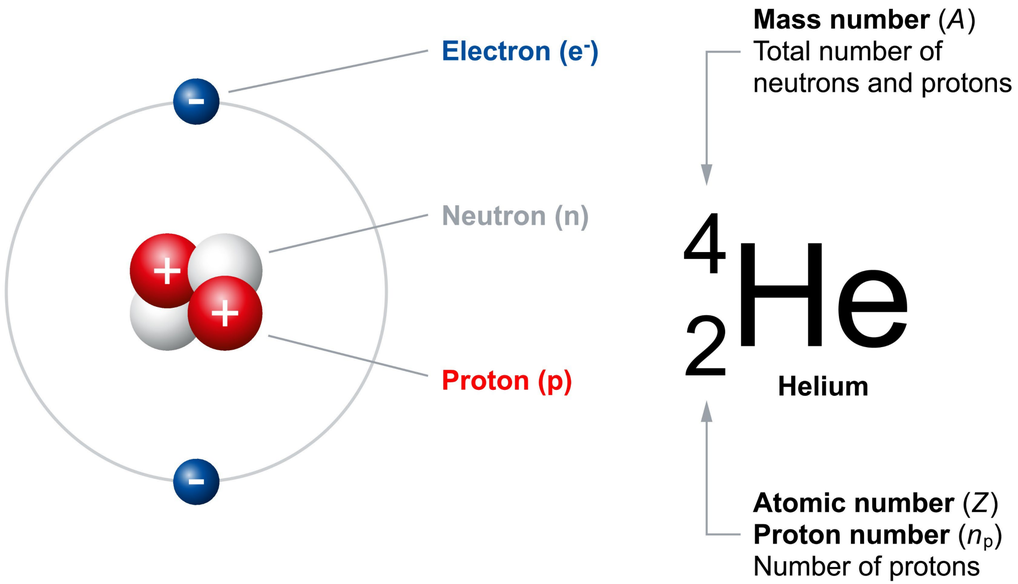
$m_{electron} \approx 1/2000\ \ m_{proton}$

→ *Electrons have virtually no effect on the atomic mass!*





Atoms: the building blocks of matter



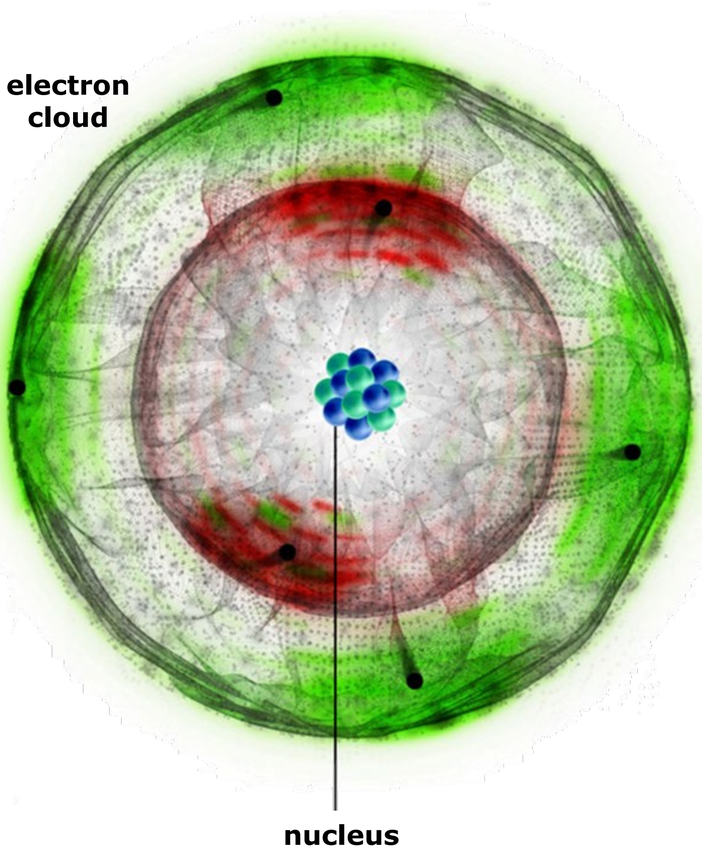
Each nucleus is a bound collection of N neutrons and Z protons. The mass number is A = N + Z, the atomic number is Z and the nucleus is written with the elemental symbol for Z.

$^{A}_{Z}[Atom]$

Mass number, A = no. protons (Z) + no. neutrons
Atomic mass = total mass of atom in amu (u)

Electrons: neutral atoms $e^- = p^+ = Z$

Atoms: the building blocks of matter



Mass number (A)
Total number of neutrons and protons

$^{4}_{2}He$

Helium

Atomic number (Z)
Proton number (n_p)
Number of protons

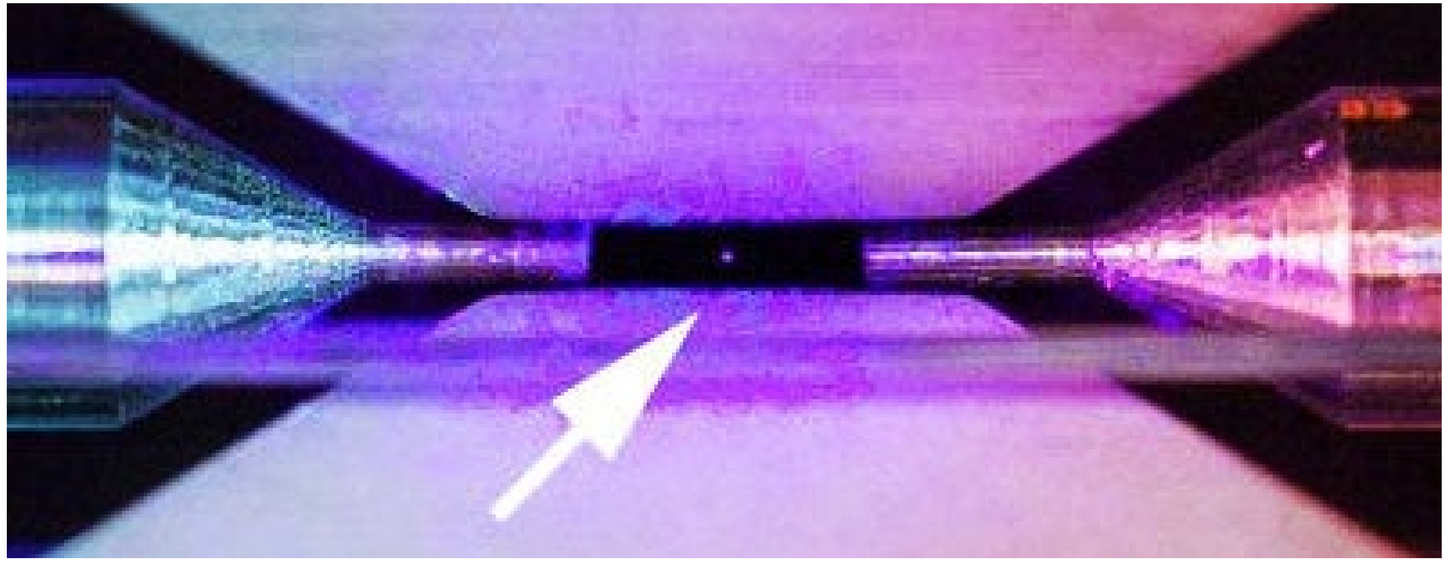
+	**proton**	positive charge (+1) 1.673×10^{-24} g = 1.0073 u atomic number, Z = no. protons
-	**electron**	negative charge (-1) 9.109×10^{-28} g = 0.0005486 u no. electrons = no. protons
n	**neutron**	electrically neutral (0) 1.675×10^{-24} g = 1.0087 u A-Z = no. neutrons

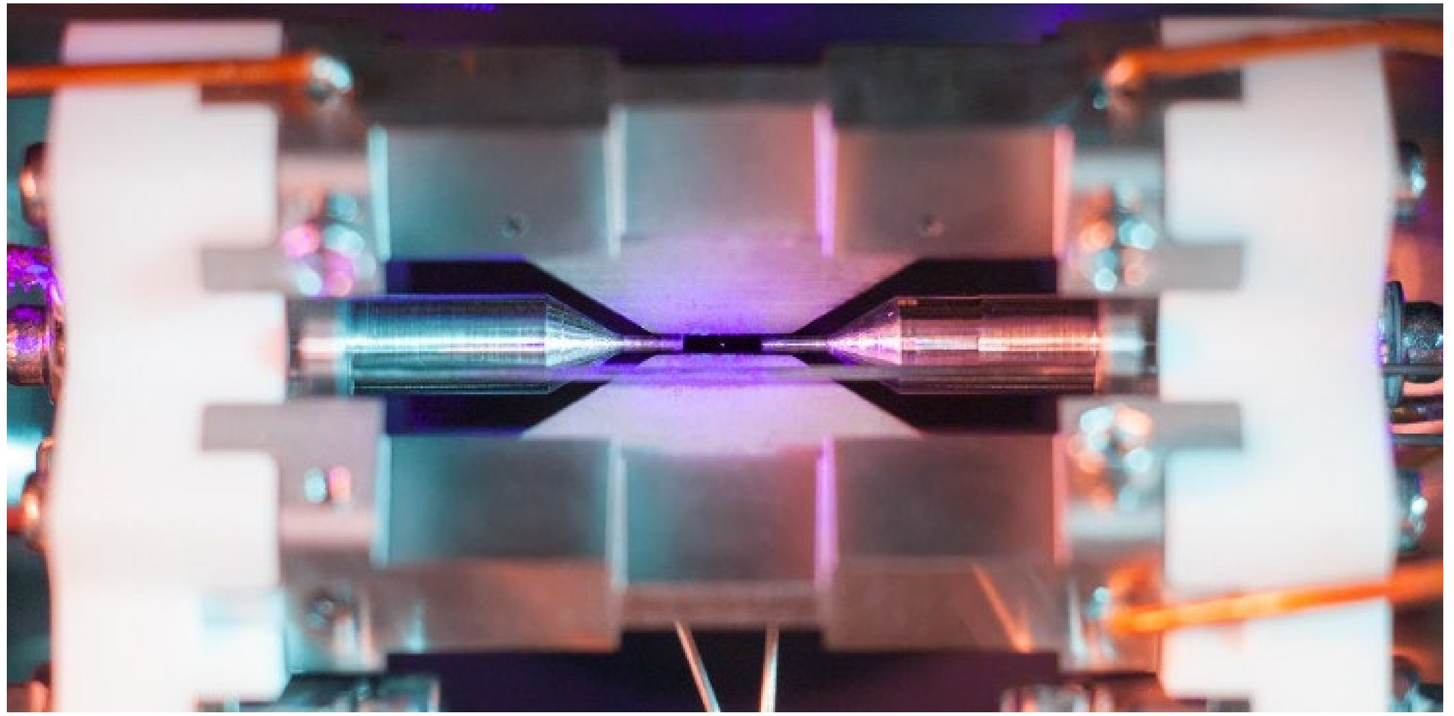
Mass number, A = no. protons (Z) + no. neutrons
Atomic mass = total mass of atom in amu (u)

11

Making an atom visible

A strontium atom in the electric field of an ion trap.





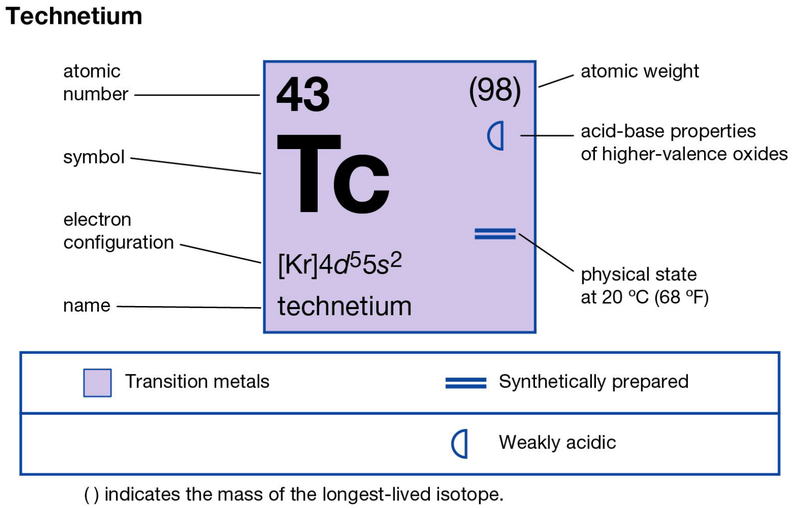
© DAVID NADLINGER VIA UNIVERSITY OF OXFORD

Chemical Elements

A chemical element is a pure substance that can no longer be broken down into other substances using chemical methods.

Mass number: number of protons + number of neutrons

Atomic number *(nuclear charge number)*: number of protons
Number of protons = number of electrons = identical chemical properties
→ Atoms with the same atomic number form an **element**.



https://pubchem.ncbi.nlm.nih.gov/periodic-table/ : Check this

Isotopes

A = no. protons (Z) + no. neutrons

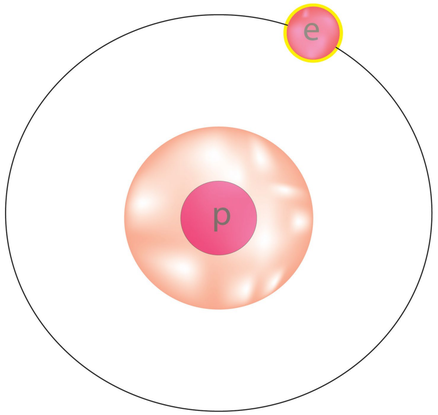
$^{A}_{Z}[Atom]$

Isotopes of an element have the **same electron shell** and therefore the **same chemical properties**.

However, due to the **different masses** of the isotopes, **reaction kinetics** are affected
→ **Isotope effect**

Special case: Hydrogen
Large **relative mass difference** between its isotopes
→ Isotopes behave **slightly differently chemically**
→ Isotopes have **their own names**

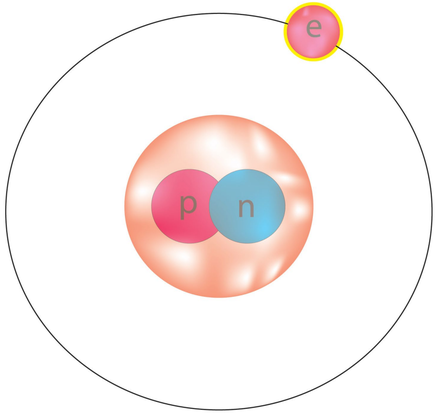
Isotopes of Hydrogen



Protium $^{1}_{1}H$

- Electrons = 1
- Protons = 1
- Neutrons = 0

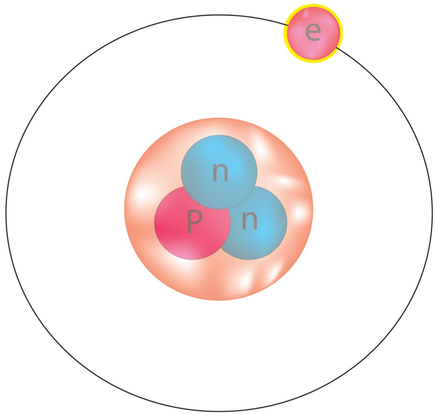
| 99.984% of natural hydrogen
| Most common form



Deuterium $^{2}_{1}H$

- Electrons = 1
- Protons = 1
- Neutrons = 1

0.0156% of terrestrial hydrogen | Known as "heavy hydrogen"; occurs mostly in HD form



Tritium $^{3}_{1}H$

- Electrons = 1
- Protons = 1
- Neutrons = 2

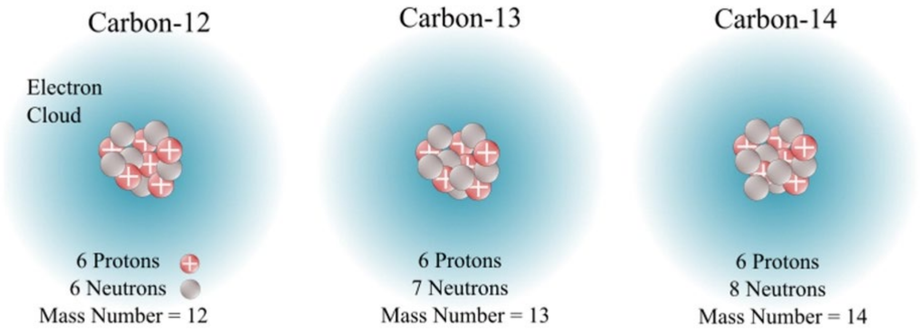
1 atom per 10^{18} atoms of protium | Radioactive; emits β^- particles (beta decay): $^{3}_{1}H \rightarrow {}^{3}_{2}He + {}^{0}_{-1}e$

Isotopes and Isobars

Same atoms of an element with a different number of neutrons.
e.g. **^{12}C, ^{13}C, ^{14}C** are isotopes of carbon all with Z= 6 and neutron

numbers N = 6, 7

$$^{A}_{Z}[Atom]$$

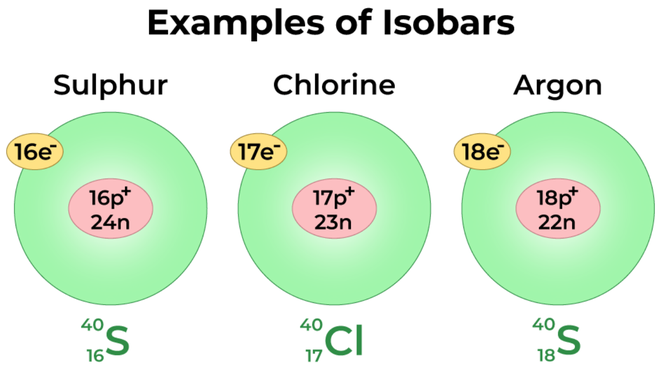


Pure element: only one stable isotope

Mixed element: two or more stable isotopes

Nuclide chart (right):
The **number of protons** in an element is always the same, the **number of neutrons** in the isotopes is variable.

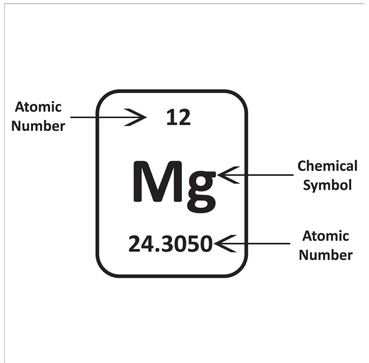
Isobars:
Nuclides with the **same mass number** but **different atomic number** (i.e. number of protons).



Atomic Mass – Review

Atomic Mass Unit (u):
Defined as **1/12 of the mass of a carbon-12 atom**.
Relative Atomic Mass:
The **ratio of the mass of an atom** to 1 atomic mass unit (u).
Absolute Atomic Mass of a Hydrogen Atom:
1.66×10^{-24} g

Relative Atomic Mass of Mixed Elements
In elements with multiple naturally occurring isotopes (e.g., Magnesium), the **relative atomic mass** is the **weighted average** of the atomic masses of all isotopes, based on their **natural abundance**



	Natural Mg Isotope Mixture	Isotope ^{24}Mg	Isotope ^{25}Mg	Isotope ^{26}Mg
Atomic Mass [u]	24.305	23.98504	24.98584	25.98259
Abundance [%]	100	78.965%	10.011%	11.025%

Exercises (Part 1): Tasks 1–9

Nuclear Reaction

Chemical reactions: → Changes in the **electron shell**, only **small amounts of energy** are released.

Nuclear reactions: → Changes in the **atomic nucleus**, energy release is approx. **10^6 times greater**.

Types of nuclear reactions:

- **1. Radioactivity** (spontaneous, natural)
- **2. Production of artificial nuclides** (bombarding nuclei with particles)
- **3. Nuclear fission** (nuclear reactor, atomic bomb)
- **4. Nuclear fusion** (hydrogen bomb)



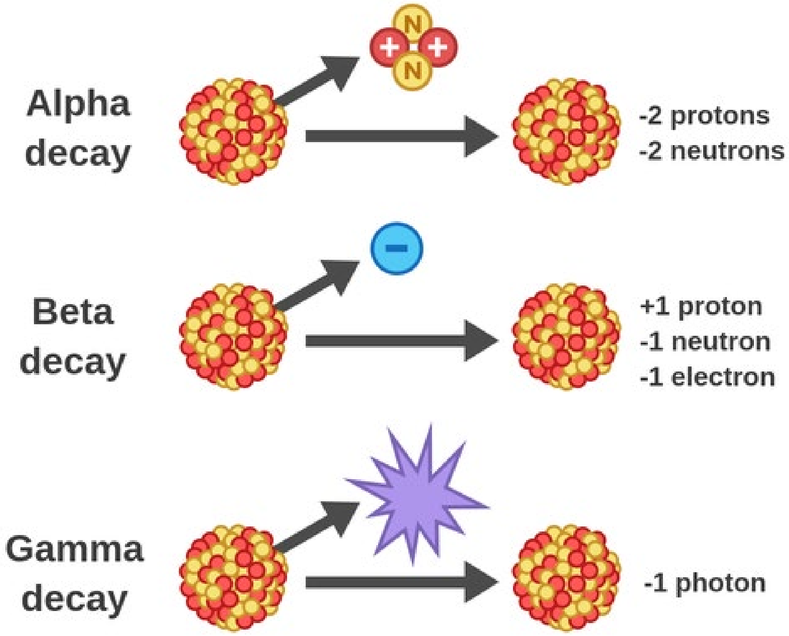
1. Radioactivity

Unstable atomic nuclei transform into **other atomic nuclei** by emitting **elementary particles** or **small nuclear fragments**.

During a **radioactive decay**, **ionizing radiation** is emitted:

- **Alpha radiation** Alpha α  $^{4}_{2}He$
- **Beta radiation** Beta β  $^{0}_{-1}e$
- **Gamma radiation** Gamma γ  energy

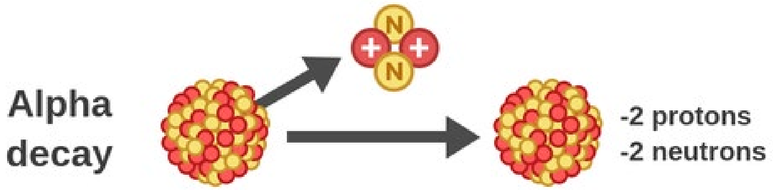
Particle	Symbols	Composition	Charge
Alpha	α	2 protons and 2 neutrons	+2
Beta	β	electron	−1
Gamma	γ	high energy electromagnetic radiations	0



Types of Decay

Alpha decay:
Emission of a **doubly positively charged helium nucleus** (*alpha particle*).
Energy is released. Alpha decay (or α-decay and alpha radioactivity) represents the disintegration of a parent nucleus to a daughter through the emission of the nucleus of a helium atom. Alpha decay is a quantum tunneling process. To be emitted, the alpha particle must penetrate a potential barrier

Alpha radiation:
Ionizing radiation consisting of **helium nuclei**, emitted by an **alpha emitter** (radioactive substance)
→ **Electrons can be removed from atoms**
→ A **stable helium atom** is formed

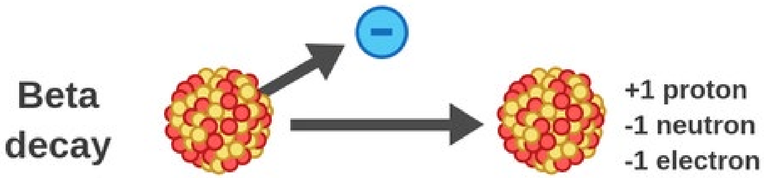


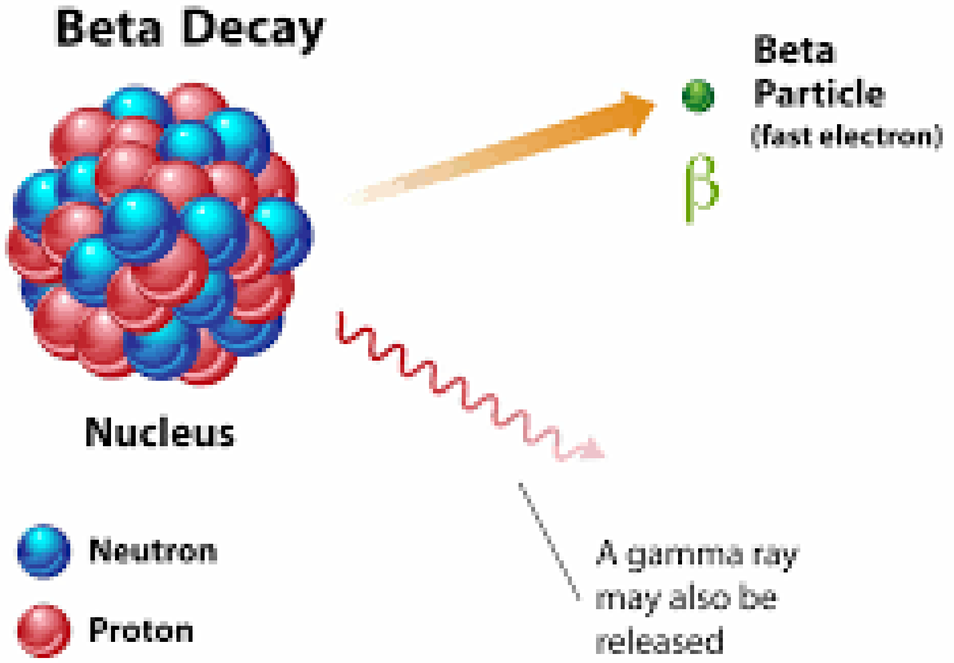
Types of Decay

Beta Decay

A **neutron in the atomic nucleus** decays into a **proton and an electron**. In the process, an **electron is emitted**.

Beta radiation consists of **electrons**.



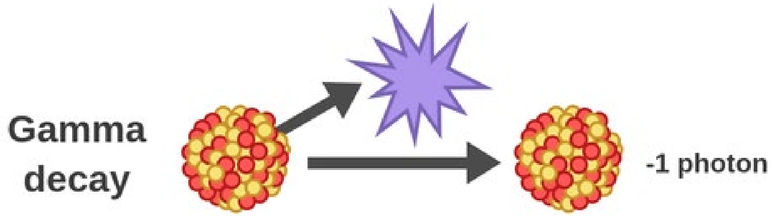


Types of Decay

Gamma Decay

The **atomic nucleus emits a high-energy photon** (= electromagnetic radiation).
Mass number and atomic number remain unchanged, only the **excitation state** of the nucleus is reduced.
→ Usually occurs **immediately after a preceding alpha or beta decay**

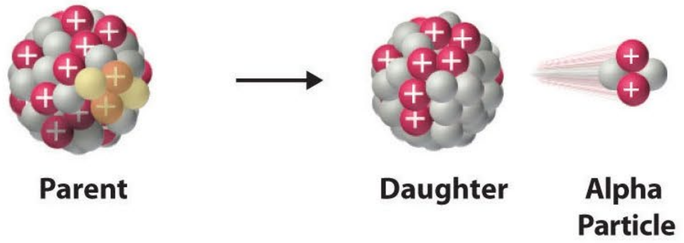
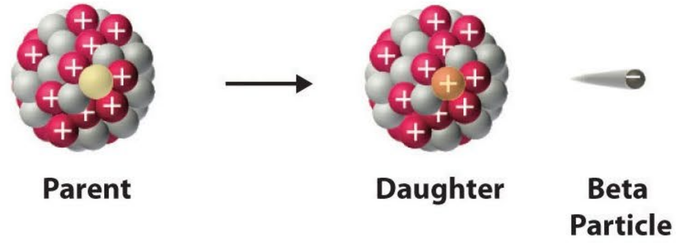
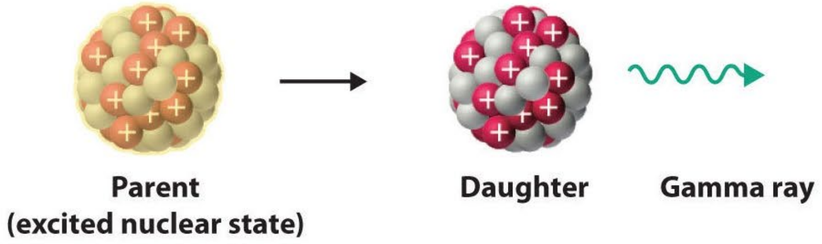
Gamma radiation:
Photons (*electromagnetic waves*)



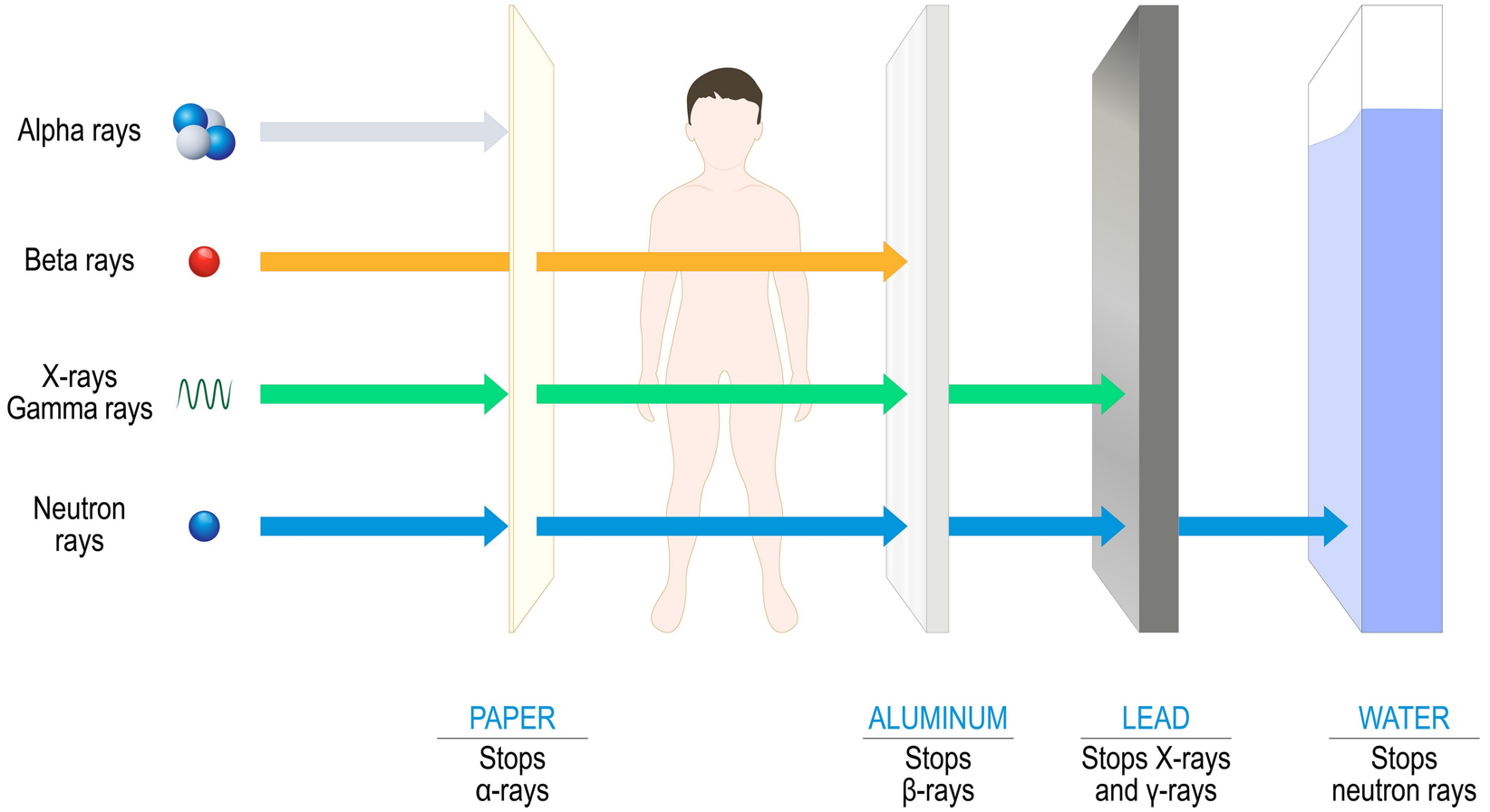
Summary of Types of Decay

Type of Radiation	Nature	Penetration / Shielding	Ionization Power	Biological Impact
Alpha (α)	Helium nucleus (2 protons + 2 neutrons)	Very low: stopped by a few cm of air or a sheet of paper	Very high	Harmless outside the body; **very dangerous if inhaled or ingested** – severe cell and tissue damage
Beta (β)	Electron (or positron for β^+)	Moderate: penetrates skin; shielded by **15 sheets of paper or ~4 mm aluminum**	Moderate	Can cause **burns on skin** and internal damage if ingested or inhaled
Gamma (γ)	High-energy electromagnetic wave	Very high: penetrates **deep into materials**; partially blocked by **lead, steel, concrete**	Low to moderate (per photon)	Can damage **internal organs** and **DNA**; high doses are **lethal**, low doses may cause **nausea**

Summary of Types of Decay

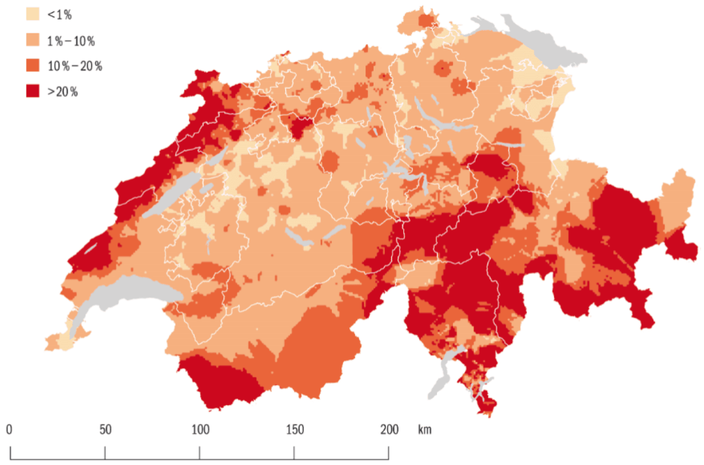
Decay Type	Radiation Emitted	Generic Equation	Model
Alpha decay	$^{4}_{2}\alpha$	$^{A}_{Z}X \longrightarrow {}^{A-4}_{Z-2}X' + {}^{4}_{2}\alpha$	
Beta decay	$^{0}_{-1}\beta$	$^{A}_{Z}X \longrightarrow {}^{A}_{Z+1}X' + {}^{0}_{-1}\beta$	
Gamma emission	$^{0}_{0}\gamma$	$^{A}_{Z}X^{*} \xrightarrow{\text{Relaxation}} {}^{A}_{Z}X' + {}^{0}_{0}\gamma$	

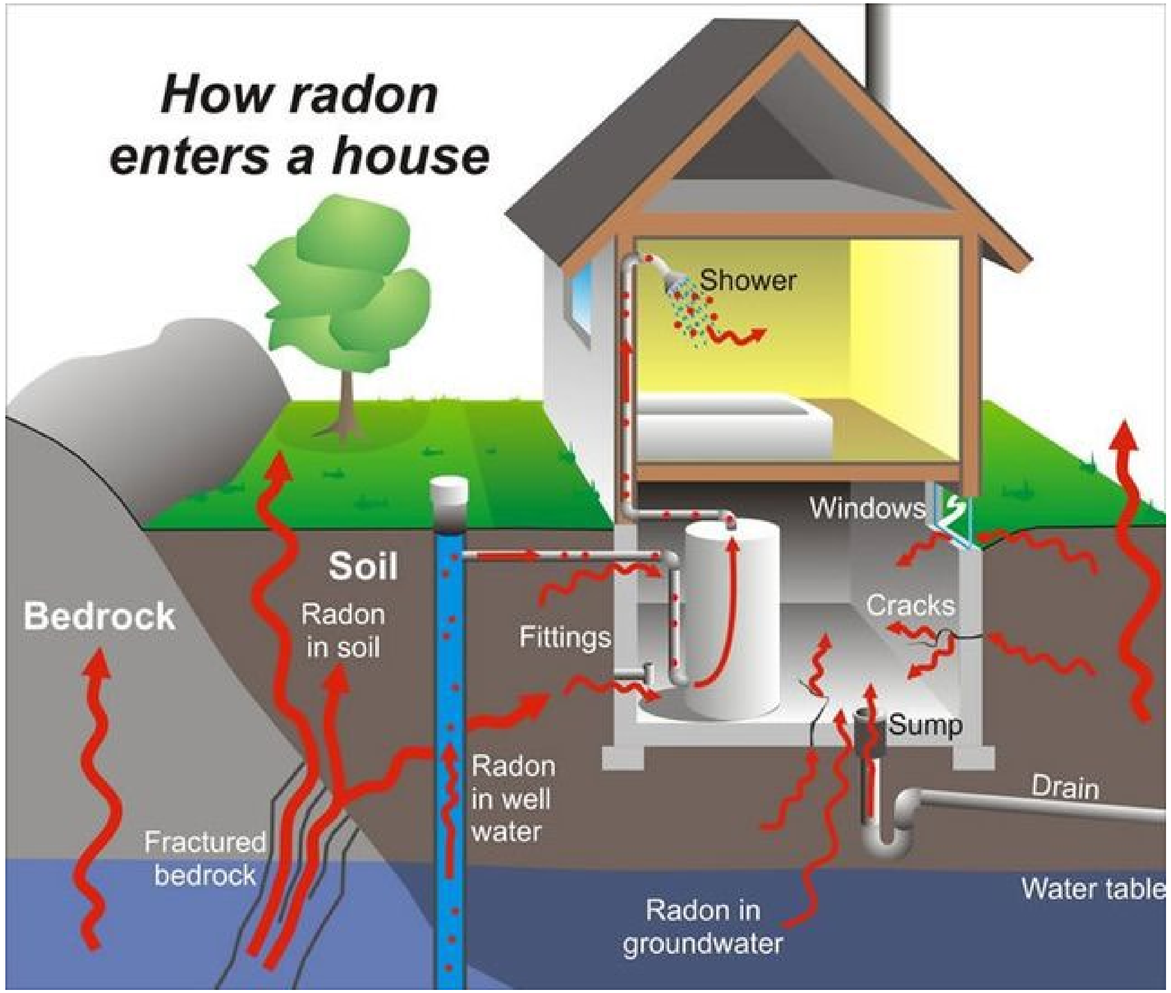
Types of radiation



Excursus: Radon

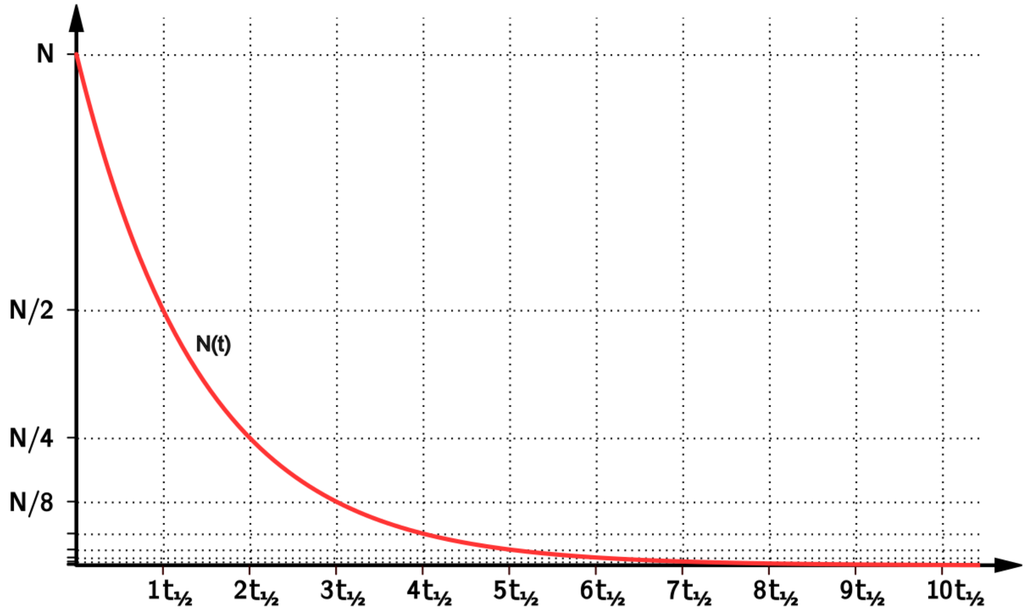
Radon is a naturally occurring **radioactive noble gas** that forms in the ground.
It can **enter buildings from the soil**, leading to **accumulation indoors**.
The **decay products** are inhaled and **settle in the lungs → lung cancer**.





Half-life ($T_{1/2}$)

The time period in which the amount, and thus the activity of a given radionuclide is reduced by half.
Radioactive decay follows an **exponential** pattern:
$T_{1/2} = \ln(2)/\lambda = 0.693/\lambda$
(λ = **natural decay constant**)



Relationship: Half-life ↔ Specific Activity

Isotope	Half-life	Specific Activity
^{131}I	8 days	4,600,000,000,000 Bq/mg
^{137}Cs	30 years	3,300,000,000 Bq/mg
^{239}Pu	24,110 years	2,307,900 Bq/mg
^{235}U	703,800,000 years	80 Bq/mg
^{238}U	4,468,000,000 years	12 Bq/mg

Why Do Atomic Nuclei Decay?

Two main factors play a role:

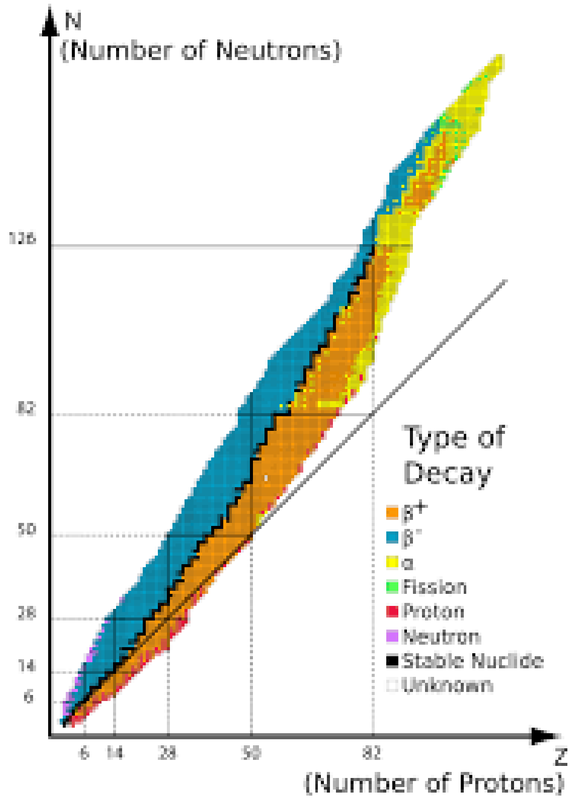
1. the **nuclear mass**
2. the **neutron-to-proton ratio**

Nuclei above a certain size (*Z* > *82*) are **all radioactive**.

In **light elements**:
Number of neutrons ≈ number of protons

In **heavy elements**:
Number of neutrons ≫ number of protons (*neutron excess*)

The **electromagnetic repulsion** between protons **exceeds** the **strong nuclear force**, which is why such nuclei are **unstable**.



https://energyeducation.ca/simulations/nuclear/nuclidechart.html

TYPE OF DECAY	MASS NUMBER (protons & neutrons)	ATOMIC NUMBER (protons)	EMITTED PARTICLE	COMMENTS
ALPHA α	-4	-2	ALPHA PARTICLE: 2 NEUTRONS + 2 PROTONS	
BETA MINUS β–	STAYS THE SAME	+1	ELECTRON	NEUTRON CHANGES INTO A PROTON, & LOSES AN ELECTRON
BETA PLUS β+	STAYS THE SAME	-1	POSITRON	PROTON CHANGES INTO A NEUTRON, & LOSES A POSITRON

Applications: Radiocarbon Dating (1)

C-14 has a half-life of ~5,730 years

Method for determining the age of carbon-containing materials (e.g., in archaeology).
→ Based on the **decay of the radioactive carbon-14 isotope (^{14}C)**.

In the atmosphere, in addition to the stable carbon isotope ^{12}C, there is also a **very small amount of radioactive ^{14}C** (approximately **one ^{14}C atom per trillion ^{12}C atoms**).

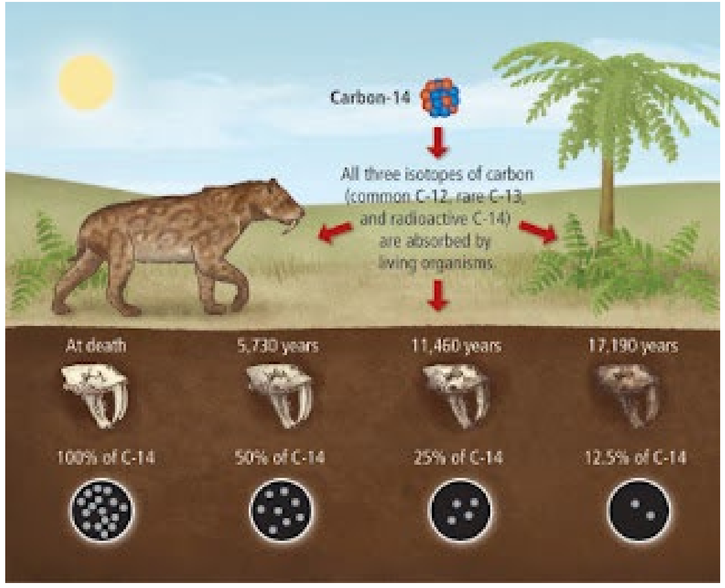
^{14}C decays with a **half-life of 5730 years but** is **constantly being regenerated** in the **upper layers of the Earth's atmosphere**.

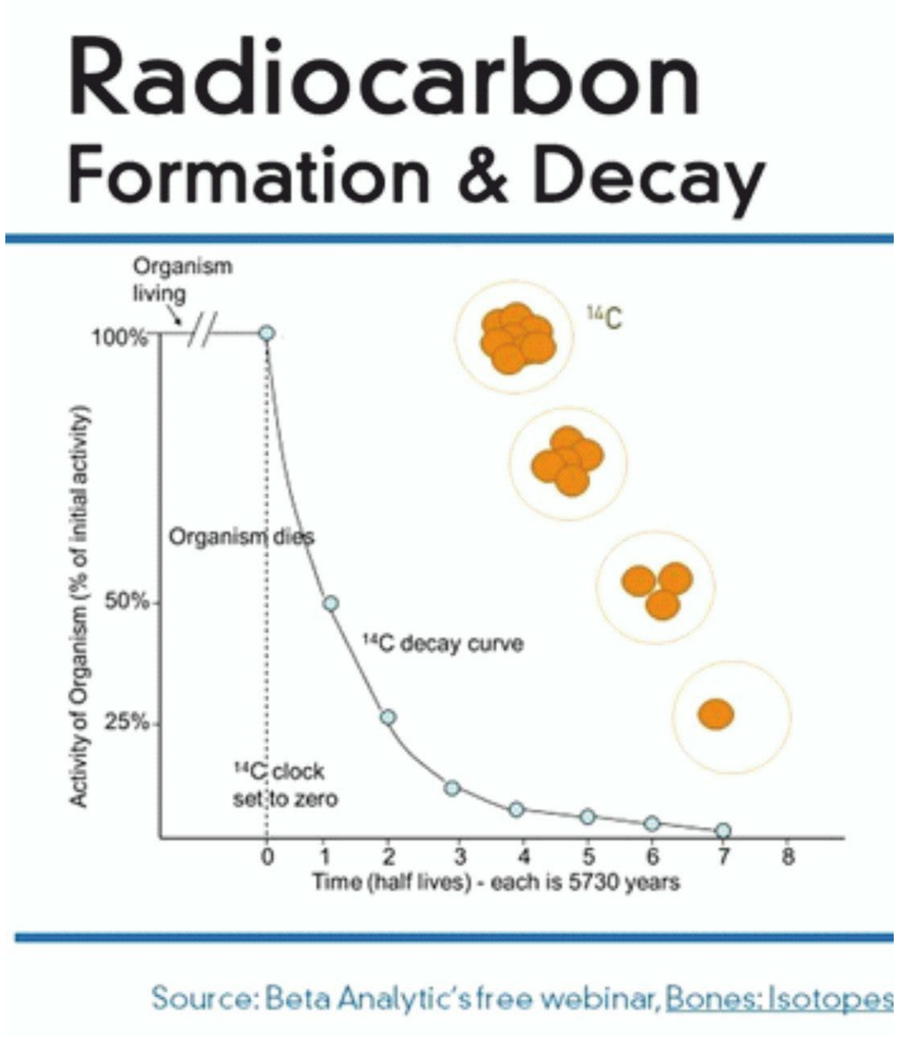
Metabolism: Living organisms maintain the **same ratio of ^{12}C to ^{14}C** as in the atmosphere.



Radiocarbon Dating (2)

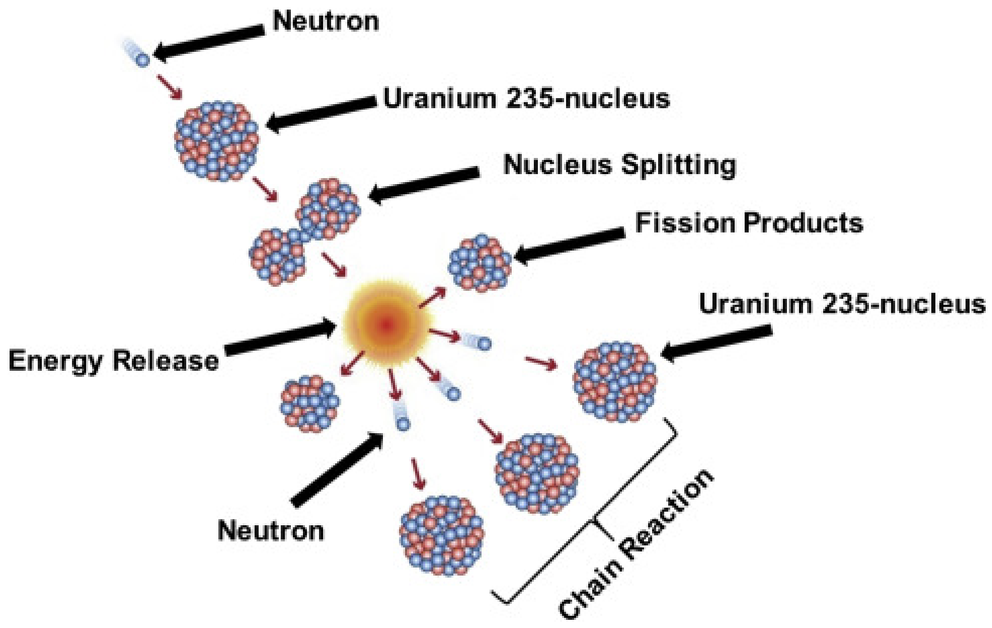
After the **death of the organism**, **no new carbon is absorbed**
→ **^{14}C decays slowly**, while **^{12}C remains stable**.
→ By **determining the $^{14}C/^{12}C$ ratio**, one can **determine the age** of a sample (wood, bone, fabric, etc.).

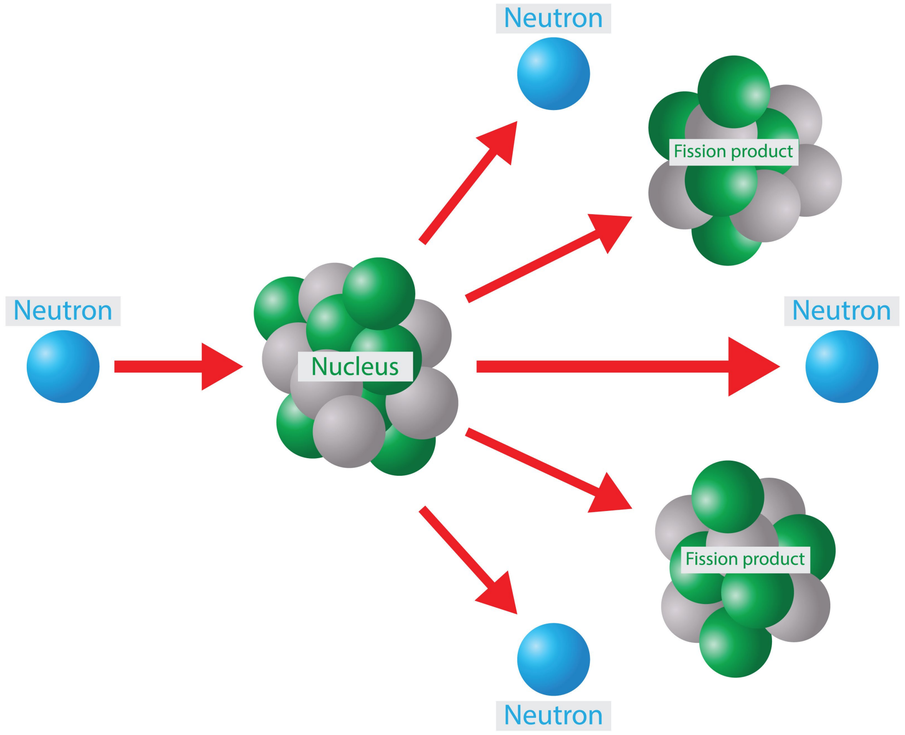




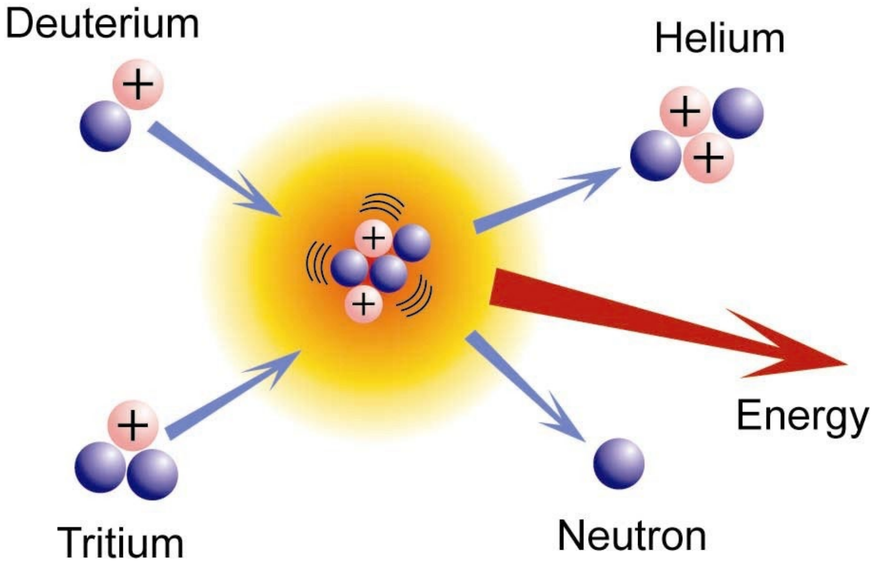
3.Nuclear Fission – The Nuclear Reactor

Nuclear fission is a reaction where the nucleus of an atom splits into two or more smaller nuclei, while releasing energy.





4. Nuclear fusion



Nuclear fusion is the process by which two light atomic nuclei combine to form a single heavier one while releasing massive amounts of energy. Nuclear Fusion reactions power the Sun and other stars.

Fusion” of two nuclei with the release of a large amount of energy.



Applications in Energy and Environmental Engineering

Field	Application Area	Radiation Use	Benefit
Energy Engineering	Nuclear Power	Fission radiation (neutrons, gamma)	Heat generation for electricity production
	Fusion Research	High-energy radiation (neutrons)	Potential clean, abundant energy
	Solar Energy	Solar radiation (photons)	Photovoltaics for electricity, CSP for heat
	Material Testing	Gamma radiography, neutron imaging	Detect defects in turbines, welds, reactor parts
	Fuel Optimization	Radiation analysis of materials	Improved reactor fuel efficiency and safety
Environmental Engineering	Water Treatment	Gamma rays, electron beams, X-rays	Disinfection of wastewater, pollutant breakdown
	Sludge Hygienization	Ionizing radiation	Pathogen reduction for safer disposal/reuse
	Air Pollution Control	Electron beam radiation	Removal of SO_2 and NOx from flue gases
	Solid Waste Management	Gamma/electron radiation	Plastic degradation, sterilization of medical waste
	Monitoring & Safety	Infrared, UV, gamma sensors	Detect leaks, monitor contamination, track pollution
Emerging Applications	Advanced Oxidation	Radiation + chemical oxidants	Enhanced breakdown of persistent pollutants
	Carbon Capture & Utilization	Radiation-driven chemical processes	Conversion of CO_2 into useful products

Exercises (Part 2): Tasks 10–11

Outlook for Next Week:

Atomic Models (self Study), Periodic Table and electronic configuration

